Homework 20: Determinants

M. Usman Rashid

BCS A, E, F, G

Definition 1. ***Definition of the Determinant of a*** $\mathbf{2 \times 2}$ ***Matrix.***

The *determinant* of the matrix

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

is $\det(A) = |A| = a_{11}a_{22} - a_{21}a_{12}$.

Definition 2. ***Minors and Cofactors of a Square Matrix.***

If A is a square matrix, then the minor M_{ij} of the entry a_{ij} is the determinant of the matrix obtained by deleting the ith row and jth column of A. The cofactor C_{ij} of the entry a_{ij} is $Cij = (-1)^{i+j}M_{ij}$.

Definition 3. ***The Determinant of a Square Matrix.***

The definition below is inductive because it uses the determinant of a square matrix of order $n-1$ to define the determinant of a square matrix of order n.

If A is a square matrix of order $n \geq 2$, then the determinant of A is the sum of the entries in the first row of A multiplied by their respective cofactors. That is,

$$\det(A) = |A| = \sum_{j=1}^{n} a_{1j}C_{1j} = a_{11}C_{11} + a_{12}C_{12} + \cdots + a_{1n}C_{1n}\,.$$

When you use this definition to evaluate a determinant, you are *expanding by cofactors in the first row*.

Theorem 1. ***Expansion by Cofactors.***

Let A be a square matrix of order n. Then the determinant of A is

$$\det(A) = |A| = \sum_{j=1}^{n} a_{ij}C_{ij} = a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in}\,, or \qquad \textit{ith row expansion}$$

$$\det(A) = |A| = \sum_{i=1}^{n} a_{ij}C_{ij} = a_{1j}C_{1j} + a_{2j}C_{2j} + \cdots + a_{nj}C_{nj}\,. \qquad \textit{jth column expansion}$$

First Alternative Method. An alternative method is commonly used to evaluate the determinant of a 3×3 matrix A. To apply this method, copy the first and second columns of A to form fourth and fifth columns. Then obtain the determinant of A by adding (or subtracting) the products of the six diagonals, as shown in the diagram below.

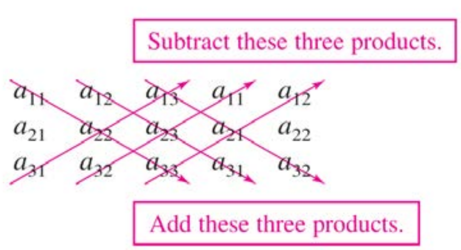

Figure 1.

Second Alternative Method. Much like we did for the 2×2 determinants, we can think of the formula for determinants of 3×3 matrices in terms of diagonals of the matrix – it is the sum of the products of its forward diagonals minus the sum of the products of its backward diagonals, with the understanding that the diagonals "loop around" the matrix (see Figure 2).

Determinant: $aei + bfg + cdh - afh - bdi - ceg$

Figure 2. A mnemonic for computing the determinant of a 3×3 matrix: we add along the forward (green) diagonals and subtract along the backward (purple) diagonals.

Problem 1. Find all the minors and cofactors of

$$A = \begin{bmatrix} 0 & 2 & 1 \\ 3 & -1 & 2 \\ 4 & 0 & 1 \end{bmatrix}.$$

Problem 2. Find all determinant of

$$A = \begin{bmatrix} 0 & 2 & 1 \\ 3 & -1 & 2 \\ 4 & 0 & 1 \end{bmatrix}.$$

Problem 3. Use *"First Alternative Method"* to find the determinant of

$$A = \begin{bmatrix} 0 & 2 & 1 \\ 3 & -1 & 2 \\ 4 & 0 & 1 \end{bmatrix}.$$

Problem 4. Use *"Second Alternative Method"* to find the determinant of

$$A = \begin{bmatrix} 0 & 2 & 1 \\ 3 & -1 & 2 \\ 4 & 0 & 1 \end{bmatrix}.$$

Problem 5. Find all determinant of

$$A = \begin{bmatrix} 1 & 2 & 3 & 0 \\ -1 & -1 & 0 & 2 \\ 0 & 2 & 0 & 3 \\ 3 & 4 & 0 & -2 \end{bmatrix}.$$

Submission Date: October 19, 2024 (Saturday – 11:59PM)

Good Luck